**STATION:** SALEM

**SYSTEM:** Rod Control

**TASK:** Take corrective actions for a dropped control rod(s)

**TASK NUMBER:** 1140330401

**JPM NUMBER:** 14-01 NRC Sim a

**ALTERNATE PATH:** X

**K/A NUMBER:** 003 AA2.03

**IMPORTANCE FACTOR:** 3.6 **RO** 3.8 **SRO**

**APPLICABILITY:**
**EO** [ ] **RO** [X] **STA** [ ] **SRO** [X]

**EVALUATION SETTING/METHOD:** Simulator

**REFERENCES:** S2.OP-AB.ROD-0002 Rev. 10 (checked 9-23-15)

**TOOLS AND EQUIPMENT:** None

**VALIDATED JPM COMPLETION TIME:** 5 minutes

**TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:** N/A

**Developed By:** **G Gauding**
**Instructor**
**Date: 08-13-15**

**Validated By:** **S Bickhart / J Militti**
**SME ~~or~~ Instructor**
**Date: 9-23-15**

**Approved By:** W Williams
**Training Department**
**Date:** 10/7/15

**Approved By:** MOB
**Operations Department**
**Date:** 10-9-15

**ACTUAL JPM COMPLETION TIME:**

**ACTUAL TIME CRITICAL COMPLETION TIME:**

**PERFORMED BY:**

**GRADE:** **SAT** **UNSAT**

**REASON, IF UNSATISFACTORY:**

**EVALUATOR'S SIGNATURE:** **DATE:**

**NAME:** ______________________

**DATE:** ______________________

**SYSTEM:** Rod Control

**TASK:** Take corrective actions for 2 dropped control rods

**TASK NUMBER:** 1140330401

**SIMULATOR SETUP** IC-241
Insert RT-1, **RD0267**, ANY ROD DROPS INTO RX, Final Severity 5 after watch has been taken.

Modify **RD0267** ANY ROD DROPS INTO RX, to Final Severity 53 when step 3.10 of S2.OP-AB.ROD-0002 is completed **OR** if Main Turbine load change is attempted due to Tavg being >1.5 degrees lower than Tref.

**INITIAL CONDITIONS:**

Salem Unit 2 is operating at 41% power, BOL.
A power reduction to bring the Main Turbine off-line is on hold.
Xe is building in @30 pcm/hr.

**INITIATING CUE:** You are the Reactor Operator. Respond to all alarms and indications.

Successful Completion Criteria:
1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:
1. Places Rod Control in Manual following 1st dropped rod.
2. Trips the reactor upon discovery of 2nd dropped rod.

NAME: ______________________

DATE: ______________________

SYSTEM: ROD CONTROL

TASK: Take corrective actions for a dropped control rod(s)

| * | STEP NO. | STEP (*Denotes a Critical Step) | STANDARD | EVAL S/U | COMMENTS (Required for UNSAT evaluation) |
|---|---|---|---|---|---|
| | | **SIMULATOR OPERATOR**:<br><br>Insert **RT-1 RD0267,** ANY ROD DROPS INTO RX, Final Severity 5, when operator assumes the watch. | Announces cluster of rod related "E" Window OHAs as unexpected alarms. Announces indications of rod 2SA1 dropped into the core.<br><br>Enters S2.OP-AB.ROD-0002, Dropped Rod. | | |
| | 2.1 | IF more than one rod is verified to be tripped, THEN Manually **TRIP** Reactor AND **GO TO** 2-EOP-TRIP-1, Reactor Trip OR Safety Injection. | Verifies only 1 rod has dropped into core. | | |
| * | 3.1 | **PLACE** Rod Bank Selector Switch in MAN. | Places Rod Bank Selector Switch in MAN.<br><br>Note: Auto outward rod movement will occur at T+ 1:30 if rods are not placed in manual. | | |
| | 3.2 | IF a Turbine load change is in progress… | Verifies no turbine load change in progress. | | |

NAME: ______________________

DATE: ______________________

SYSTEM: ROD CONTROL

TASK: Take corrective actions for a dropped control rod(s)

| * | STEP NO. | STEP (*Denotes a Critical Step) | STANDARD | EVAL S/U | COMMENTS (Required for UNSAT evaluation) |
|---|---|---|---|---|---|
| | 3.3 | **ADJUST** $T_{ave}$ to within $1.5^0$ of program as follows:<br>• <u>IF</u> Main Turbine is operating, <u>THEN</u> adjust Turbine load.<br>• <u>IF</u> Main Turbine is NOT operating, <u>THEN</u> **ADJUST** Steam Dumps OR 21-24MS10 valves. | Verifies $T_{ave}$ is within $1.5^0$ of program.<br><br>**Simulator Operator: <u>IF</u> attempt to lower turbine load is started, then change RT-1 as described on next page <u>now</u>.** | | |
| | 3.4 | Is Reactor subcritical as a result of the dropped rod? | Answers NO, <u>GOES TO</u> step 3.9 | | |
| | 3.9 | <u>IF AT ANY TIME</u> a power reduction becomes necessary, <u>THEN</u> **BORATE** <u>AND</u> **ADJUST** Turbine load or Steam Dump System flowrate to maintain $T_{ave}$ within $1.5^0$ F of program. | Determines no power reduction is necessary. | | |
| | 3.10 | Is power above 50% of RATED THERMAL POWER? | Answers NO, and GOES TO Step 3.12 | | |
| | | | **<u>SIMULATOR OPERATOR</u>**:<br>Modify malfunction **RD0267,** ANY ROD DROPS INTO RX, to Final Severity 53, when step 3.10 of S2.OP-AB.ROD-0002 is completed. | | |

NAME: ______________________
DATE: ______________________

SYSTEM: ROD CONTROL

TASK: Take corrective actions for a dropped control rod(s)

| * | STEP NO. | STEP (*Denotes a Critical Step) | STANDARD | EVAL S/U | COMMENTS (Required for UNSAT evaluation) |
|---|---|---|---|---|---|
| * | 2.1 | <u>IF</u> more than one rod is verified to be tripped, <u>THEN</u> Manually **TRIP** Reactor <u>AND</u> **GO TO** 2-EOP-TRIP-1, Reactor Trip OR Safety Injection. | Observes a second rod bottom light by direct observation, or by depressed power in the region of the 2 dropped rods, Terr or Tavg lowering , or OHA D-32 TAVE LO, and MANUALLY **TRIP** the Reactor IAW Step 2.1.<br>**Evaluator:** See next step if second dropped rod is not recognized and procedure is continued (This will allow continuation in the JPM until the 2 times validation time has been reached and the JPM is terminated.) | | |
| | 3.12 | **REQUEST** Maintenance to determine if an Individual Rod Position Indicator (IRPI) malfunction has occurred. | Contacts Maintenance or requests CRS to contact Maintenance to determine if an IRPI malfunction has occurred.<br>**Cue:** Maintenance has been contacted. | | |
| | 3.13 | Has an IRPI malfunction occurred? | Answers NO based on rod bottom, OHAs, and primary plant parameter changes, and GOES TO Step 3.15. | | |
| | 3.15 | **INITIATE** a power reduction to <75% Rated Thermal Power… | Recognizes power is 40%. | | |

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DATE: ______________________

SYSTEM: ROD CONTROL
TASK: Take corrective actions for a dropped control rod(s)

| * | STEP NO. | STEP (*Denotes a Critical Step) | STANDARD | EVAL S/U | COMMENTS (Required for UNSAT evaluation) |
|---|---|---|---|---|---|
| | 3.16 | **REQUEST** Reactor Engineering assistance to recover rod. | Contacts Reactor Engineering or requests CRS to contact Reactor Engineering for assistance in recovering dropped rod.<br>**Cue:** Reactor Engineering has been contacted. | | |
| | 3.17 | Is dropped rod to be recovered, per Reactor Engineering? | **Cue:** Reactor Engineering will be performing a flux map to aid in determination of whether a recovery will be made of the dropped rod. | | |
| | | | Terminate the JPM once a manual Reactor Trip has been performed or time reaches two times the validation time. | | |

## JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

**NOTE:** All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

S a 1. Task description and number, JPM description and number are identified.

S a 2. Knowledge and Abilities (K/A) references are included.

S a 3. Performance location specified. (in-plant, control room, or simulator)

S a 4. Initial setup conditions are identified.

S a 5. Initiating and terminating Cues are properly identified.

S a 6. Task standards identified and verified by SME review.

S a 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

S a 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 10 Date 12-5-07

S a 9. Pilot test the JPM:
   a. verify Cues both verbal and visual are free of conflict, and
   b. ensure performance time is accurate.

________ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

________ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [signature] Date: 9-23-15

SME/Instructor: [signature] BICKFORD. Date: 9/23/15

SME/Instructor: ________ Date: ________

**INITIAL CONDITIONS:**

Salem Unit 2 is operating at 41% power, BOL.
A power reduction to bring the Main Turbine off-line is on hold.
Xe is building in @30 pcm/hr.

**INITIATING CUE:**

You are the Reactor Operator. Respond to all alarms and indications.